Synchronous and Asynchronous Communication

In modern architectures, services are constantly talking to each other —
Payment Service, Order Service, Notification Service, Inventory Service...

The way they talk defines your system's speed, reliability, and scalability.

There are two ways these services can communicate —
Synchronous (wait for response) and Asynchronous (fire and forget).

Synchronous Communication

-> Client sends a request.

-> Waits until the server finishes and sends back a response.

-> Only then can the client move forward.

Example in real life:
You call a restaurant and wait on
the phone until they confirm your order.

Advantages:

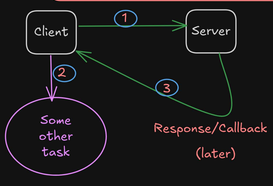
-> Simple and predictable
-> Real-time feedback (user gets result immediately)
-> Easier to debug and trace

Disadvantages:

-> Slow if server takes time
-> If one service fails → whole chain breaks
-> Poor scalability for large systems

DEMO: Synchronous Communication

Asynchronous Communication



-> Client sends a request but doesn't wait.
-> Server processes it later.
-> Response (if any) comes later through:
--> Callback / webhook
--> Polling
--> Message queue (we'll cover this in next video)

Example in real life:
You order food on Zomato —
you get "Order Confirmed" instantly,
and food arrives later.

Advantages:

-> Fast response for user (non-blocking)
-> Can handle high load easily
-> More resilient — if one part is slow, others keep working

Disadvantages:

-> Harder to debug
-> Responses may come late
-> Data may be eventually consistent, not immediate

DEMO: Asynchronous Communication

When to Use Which?

Use Synchronous when:

You need immediate response

Action must happen in sequence

Small, low-latency operation

Use Asynchronous when:

You have background or delayed tasks

You want to handle large traffic bursts

You need to decouple services

Synchronous Examples:

REST API → Client waits for JSON response

gRPC call → Immediate result

Asynchronous Examples:

Message Queue (RabbitMQ, Kafka, SQS)

Webhook callbacks

Background jobs / worker threads

Push notifications